

JOSHUA HARTY

Software Engineer — High-Performance Python / C++ · Risk & Pricing · Agentic Systems

London, UK · joshh456@icloud.com · [My Site](#)
linkedin.com/in/joshua-h-b5aa3a24b · github.com/slimboi34

Software engineer building high-performance pricing and risk systems in the London specialty insurance market. Author of **RiskPY**, an open-source C++/Python actuarial pricing framework on PyPI. Ships production LLM-backed features and serverless AWS services, and runs an AI-first development workflow — coding agents and LLM automation across projects and the delivery cycle. Pursuing an MS in Computer Science (CU Boulder) alongside full-time work.

EXPERIENCE

Software Engineer · Liberty Specialty Markets

Mar 2025 – Present

London, UK (Hybrid) — specialty insurance: risk & pricing

- Built an **admin panel over the risk-pricing models** giving stakeholders per-policy insight, trend analysis across the underwriting mix, and system-wide data exportability — including a **natural-language query interface** so non-technical stakeholders can self-serve insights.
- Designed an **agentic workflow** automating non-critical dev changes end-to-end through Jira, built on the team's existing agentic API and running in production.
- Built a harness connecting internal agents to **Slack and Microsoft Teams** — production tooling in active internal use, which I maintain and am extending with a developer bot.
- Run an **AI-first dev flow**: coding agents and LLM automation embedded throughout day-to-day delivery, from code generation to routine engineering work.
- Design and ship **serverless services on AWS** — Lambda, Step Functions, API Gateway — deployed with CDK and GitHub Actions CI/CD, with CloudWatch observability.

System Integration Engineer · Light & Wonder

Sep 2023 – Mar 2025

Chiswick, London (Hybrid) — gaming systems

- Built and maintained **Python integration and test automation** for regulated gaming platforms, with Jenkins CI pipelines across multi-system environments.

Earlier roles · Paragon Impact (Jersey) · The Good People Data Company (Johannesburg)

Oct 2022 – Sep 2023

- Junior engineer roles: Python back-end development — JSON APIs and data validation for an ESG reporting product; client data products and front-end contributions.

OPEN SOURCE & PROJECTS

RiskPY · creator & maintainer — `pip install open-riskpy`

PyPI · GitHub · MIT

- Open-source **actuarial pricing framework** bridging native C++ performance with a Python API via pybind11 and scikit-build-core.
- Monte Carlo aggregate-loss simulation (Poisson × Lognormal) in native C++; chain-ladder reserving with IBNR; Bühlmann credibility; excess-layer pricing, ILFs, burning-cost analysis; rate-filing tools.
- Automated release: GitHub Actions CI/CD publishing to PyPI via **Trusted Publishers**; ships a desktop GUI and batch CSV-to-Excel processing.

Selected Personal Projects · github.com/slimboi34

105 repositories

- Agentic systems** — *agent-ventures*: eight deployable micro-SaaS products on one shared zero-dependency kit, Stripe-wired and Railway-deployed. Plus an agentic Telegram bot driving a personal LLM workflow end-to-end, and a self-resuming autonomous task runner.
- Quant & markets** — Polymarket prediction-market paper-trading and calibration analytics (Python); an order-flow tool surfacing what the book does beneath top-of-book; a Monte Carlo risk visualiser.
- Systems & graphics** — C++ 360° tower-defence game; Rust/Bevy Apollo 11 landing simulation; this CV site — a WebGL particle field of 120k GPU particles with a live in-browser Monte Carlo loss simulation.
- Data & ML** — NLP disaster-tweets classifier (Kaggle); interactive UK homicide geospatial map; Solidity on-chain wagering contracts and Zama FHE experiments.

EDUCATION

MS, Computer Science · University of Colorado Boulder

In progress · exp. 2027

- Coursework: advanced algorithms & data structures, dynamic programming, approximation & linear programming, high-performance & parallel computing, deep learning (vision & NLP), data mining, big-data architecture, FPGA & real-time embedded systems.

BSc, Information Technology · Richfield

2022 – 2024

SKILLS

Languages	Python, C++, TypeScript/JavaScript, SQL, Rust (hobbyist)
Cloud & Infra	AWS (Lambda, Step Functions, API Gateway, CDK/SAM, CloudWatch), GitHub Actions, Jenkins, Docker, Railway
Domains	Actuarial pricing & risk, Monte Carlo simulation, LLM/agent engineering, MCP tooling, AI-augmented development, pybind11 native extensions, Django/React
Spoken	English (native), Afrikaans (native)

HONOURS

4th in the world World Rowing Junior Championships — junior men's coxed four, representing South Africa